

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Forecasting SAM 3's reliability on unseen species & places — from label-free
distances

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The problem

Promptable trackers (**SAM 3**) segment & track *any* named animal in a video with **no training on that species**.

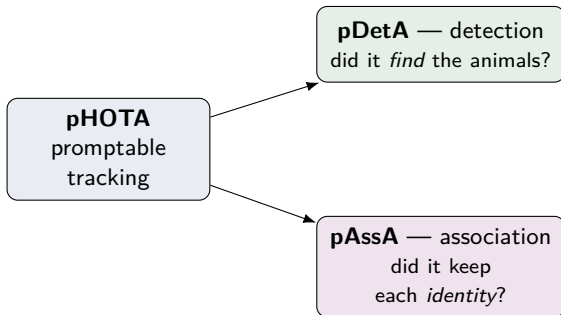
- Conservation teams point them at new species / regions **constantly**.
- But nobody can say *in advance* whether the model is **trustworthy** for a given animal in a given place.
- There is **no rule** that predicts this. That gap is what we fill.

The question

Can we predict SAM 3's zero-shot accuracy on an **unseen (species, place)** *before running it* — from properties measurable in advance?

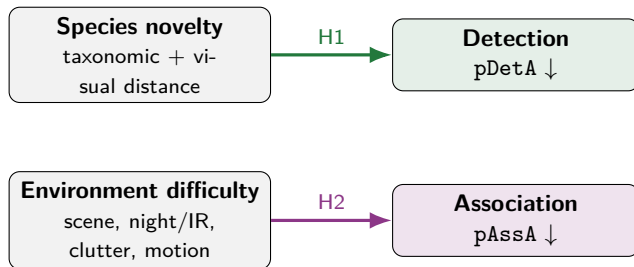
- **Not** trying to make the tracker score higher — we predict & explain its reliability.
- And: *which factors* govern transfer, separately for **finding** the animal and **following** it?

The metric decomposes — and that is the point



We analyse the **two components separately** (HOTA-style) — the hypothesis is they transfer for *different reasons*.

Two hypotheses



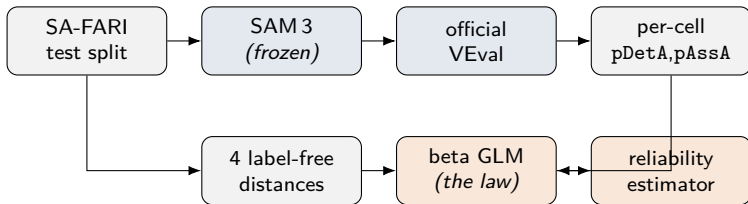
H0 (still a result): if distance explains little → transfer isn't distance-driven; pivot to representational probing.

The data — SA-FARI: a natural experiment

- Largest open multi-animal wildlife tracking set (Meta $\times$ Conservation X Labs).
- **99 species** (full 7-level taxonomy) · **741 locations** / 4 continents · a decade of footage.
- **Train / test disjoint by species *and* location** $\Rightarrow$ every test video is a *controlled probe* of transfer along a measurable distance.
- Hard negatives (query returns nothing) are **kept** — they drive false-positive analysis.

Species $\times$ place $\times$ time = the axes we measure distance along.

Approach — freeze the model, fit a small law



SAM 3 **frozen**; the metric is the **official** one. Only the small **beta GLM** is fit — from label-free distances. **No target label enters a distance.**

The four label-free distances (+ a proxy)

- **Taxonomic** — tree steps to the nearest training species (lowest common ancestor).
- **Visual** — DINOv2 embedding of the *mask-cropped* animal vs. nearest training prototype (CLIP variant as a robustness check).
- **Environment** — scene embedding of the masked-out background vs. nearest training location; + a night/IR flag.
- **Temporal** — years to the nearest relevant training footage.
- **SAM-3 familiarity proxy** — separability in SAM 3's *own* feature space (hedges the unknown pretraining set).

All computed against the **frozen training split only** — the leakage firewall.

Novelty — stated honestly

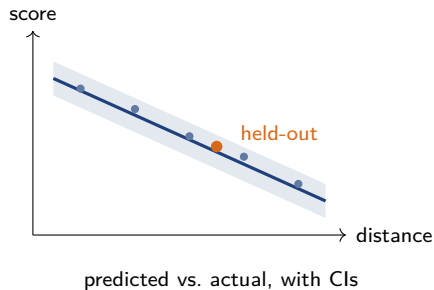
Not new: that OOD performance is predictable (for *classifiers*: Accuracy- & Agreement-on-the-Line); taxonomy in embeddings (BioCLIP); location shift (Terra Incognita).

Novel — the combination

First predictable-transfer study for a **promptable video tracker, decomposed into two predictors** (detection vs association), on SA-FARI's species $\times$ place $\times$ time natural experiment — a deployment “**will it work here?**” estimator.

Validation & honesty

- Hold out **whole species & whole locations** (group-aware CV).
- Predict pDetA/pAssA from *distance alone*; report **bootstrap CIs** everywhere.
- **Success** = a *validated* predictor, not a higher score.
- **Gate**: no out-of-sample signal $\rightarrow$ invoke H_0 , pivot.



Plan & contribution

- 1 Setup; freeze the seen/unseen reference (leakage firewall).
- 2 Frozen SAM3 inference $\rightarrow$ VEval $\rightarrow$ per-cell scores. [gate: scores sane?]
- 3 Build the four distances + proxy; assemble the feature table.
- 4 Fit per-target beta GLMs; variance partition; decomposition test.
- 5 Grouped CV; bootstrap; predictive-line figures. [gate: real signal? else H0]
- 6 Ablations & robustness; reliability estimator; write-up.

The deliverable: a **validated, label-free reliability estimator** for a promptable tracker — and *why* detection and association transfer differently.

Thank you — questions welcome.